

Task 1 – VPC Creation (6 Marks)

VPC dashboard
EC2 Global View

Filter by VPC:
Select a VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

VPCs (1)

VPC	Name	CIDR block	State	IPv4 CIDR
☒	vpc-0a1b2c3d4e5f6g7h	10.10.0.0/16	Available	10.10.0.0/16

Subnets (2)

Subnet	Subnet ID	VPC	CIDR block	Availability Zone	Auto-assign public IPv4
☒	GLA-Yashansh-Public-1	subnet-0a1b2c3d4e5f6g7h	10.10.1.0/24	ap-south-1a	Yes
☒	GLA-Yashansh-Public-2	subnet-0b2c3d4e5f6g7h	10.10.2.0/24	ap-south-1b	Yes

Internet gateways (1)

Internet gateway	Name	ID	State
☒	GLA-Yashansh-IGW	igw-0a1b2c3d4e5f6g7h	Attached

Route tables (1)

Route table	Name	ID	State
☒	GLA-Yashansh-Public-RT	rtb-0a1b2c3d4e5f6g7h	Associated

Routes

Destination	Target
0.0.0.0/0	igw-0a1b2c3d4e5f6g7h

Task 2 – IAM Configuration (6 Marks)

IAM

User groups (1)

Group name	Attached policies	Creation time
☒ GLA-Notice-Developers	1	November 25, 2024, 12:10 (UTC+05:30)

Users (1)

User name	Groups	Last activity	Creation time
☒ GLA-Yashansh-User	GLA-Notice-Developers		November 25, 2024, 12:12 (UTC+05:30)

Permissions attached to GLA-Notice-Developers (1)

Policy	Type	Description
☒ AmazonEC2FullAccess	AWS managed	Provides full access to Amazon EC2 via the AWS Management Console

Task 3 – EC2 Deployment (6 Marks)

Instances (1)

Name	Instance ID	Instance state	Instance type	Public IPv4 address
☒ GLA-Yashansh-Notice-Server	i-0a1b2c3d4e5f6g7h8	Running	t2.micro	13.234.567.89

Security Groups (1)

Name	ID
☒ GLA-Yashansh-SG	sg-0a1b2c3d4e5f6g7h

Inbound rules

Type	Protocol	Port range	Source
SSH	TCP	22	0.0.0.0/0
HTTP	TCP	80	0.0.0.0/0

```
[ec2-user@ip-10-10-1-10 ~]$ sudo yum update -y
Complete!
[ec2-user@ip-10-10-1-10 ~]$ sudo yum install httpd -y
Complete!
[ec2-user@ip-10-10-1-10 ~]$ sudo systemctl start httpd
[ec2-user@ip-10-10-1-10 ~]$ sudo systemctl enable httpd
Created symlink /usr/share/systemd/systemd-sysv-generator: httpd.service.
[ec2-user@ip-10-10-1-10 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
Active: active (running)
```

Task 4 – Webpage Hosting (6 Marks)

Welcome to GLA Department Notice Portal

Student Name: Yashansh Srivastava
Roll Number: 2415001905

Task 5 – S3 and CloudWatch (6 Marks)

S3 Bucket

Buckets (1)

Name	AWS Region	Creation date
☒ gla-yashansh-notice-resources	Asia Pacific (Mumbai)	November 25, 2024, 12:30 (UTC+05:30)

Objects (1)

Item	Type	Size	Last modified
☒ department-notice.pdf	pdf	124.3 KB	November 25, 2024, 12:32 (UTC+05:30)

CloudWatch Metric

Metrics (2 selected)

Metric	Unit	Stat
☒ CPUUtilization	Percent	Average
☒ NetworkIn	Bytes	Average

CloudWatch Alarm

Alarms (1)

Name	State	Metric	Threshold
☒ GLA-Yashansh-CPU-Alarm	In alarm	CPUUtilization	> 60% for 5 minutes

CPUUtilization

Task 6 – Validation and Demonstration (6 Marks)

- ✓ VPC (GLA-Yashansh-Notice-VPC with 2 public subnets and IGW)
- ✓ IAM Group and User (GLA-Notice-Developers, GLA-Yashansh-User)
- ✓ EC2 Instance (running with Apache, SSH & HTTP access)
- ✓ Hosted Webpage (accessible using public IPv4 address)
- ✓ S3 Bucket (with uploaded department-related file)
- ✓ CloudWatch Metrics and Alarm (CPUUtilization and NetworkIn)

Task 7 – Explanation (6 Marks)

- **Application Load Balancer (ALB):**
 - Distributes incoming traffic across multiple EC2 instances.
 - Improves availability and fault tolerance.
 - Automatically handles instance failures.
 - Supports health checks and scalability.
- **CloudFront:**
 - Content Delivery Network (CDN) that caches content at global edge locations.
 - Reduces latency for users across different regions.
 - Improves performance and availability of the portal.
 - Works with ALB/S3 to deliver static and dynamic content efficiently.